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(1) Research Program/Project/Study Title: Healing Effect of Ethnobotanical Extracts on Wounds of Capon Darag Native Chicken Cockerel (<i>Gallus gallus domesticus</i>)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological/Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Dumarao						
		Province:	Capiz						
		Region:	6						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								

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(7) Sector/Commodity Livestock	(8) Discipline/Field of Specialization Animal Science
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(9A) Abstract

The aim of this research is to determine the healing effect of different ethnobotanicals (Turmeric, *Mayana* and *Hagonoy*) on the wounds of capon Darag native chicken cockerel. This study used a Completely Randomized Design (CRD), which is divided into five treatments with three replications. The treatments of the study are the following: T1 – Turmeric (*Curcuma longa*) plant extracts, T2 – *Mayana* (*Plectranthus scutellariodes* L.) plant extracts, T3 – *Hagonoy* (*Chromolaena odorata* L.) plants extracts, T4 – Combined Turmeric, *Mayana* and *Hagonoy* plant extracts and T5 – Control (Betadine). Darag native chicken cockerels were used in this study. Incisions of 2cm per cockerel were done to expose and remove the two testes. The wounds were then sprayed with the different plant extracts and commercial wound solution for seven days. Data gathering was done for fifteen days and measurement of the wound diameter were done on the 3rd, 6th, 9th, 12th and 15th day post capon operation. The results revealed that all plant extracts were effective in the healing of wounds in capon cockerels. *Hagonoy* and *Mayana* extracts attained the earliest complete healing process of wounds at day 12 of post capon operation. All plant extracts were comparable to commercial wound solution in the healing of wounds of Darag native chicken cockerels.

(9B) Keywords : Extract, *Hagonoy*, *Mayana*, Turmeric

(10) Introduction

Capons are male chickens whose testes have been surgically removed. Because of the resultant deficiency of androgen production, caponized cockerels lack the characteristic of large comb and wattles associated with sexually mature roosters (Rikimaru *et al.*, 2011). In capons, the energy that is usually expended in fighting, courting behavior, and territorial protection is greatly reduced, thereby allowing a highly efficient conversion of feed into growth, fat deposition, and improved meat quality (Jacob and Mather, 2000). The meat from a rooster can be very stringy and pungent, but the meat from the capon is much tender (Amorim *et al.*, 2016), juicier, tastier (Rikimaru *et al.*, 2011) and best suited for roasting (inasal).

Capons have been produced from heavy breeds such as Jersey Giants, Brahmas, Orpingtons, Cornish, Plymouth Rocks, Cochins, or Cornish × Plymouth Rock cross that are typically used by the commercial broiler industry (Jacob and Mather, 2000). However, recent studies have been performed on capons of slow-growing local chickens rather than broilers because capons from heavy breeds weigh 5 or 6 kg, which exceeds the commercial weight for home consumption, and consumers demand better quality of meat (Chen *et al.*, 2000).

In caponization, a wound will be created. If not properly cleaned, this would lead to infection that might result to death of chickens. Wound healing is a challenging clinical problem and management of wounds is important (Gandu, 2017). Therefore, several commercial wound treatments are used. However, this commercial wound treatment are high cost. To solve this high cost of commercial wound solutions, an alternative treatment will be used. These alternative treatments are the use of some herbal medicines or ethnobotanicals which can be

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found anywhere. Some of these herbal medicines that have wound healing property are Turmeric, *Mayana* and *Hagonoy*.

Curcuma longa L. is known as turmeric and is belonging to Zingiberaceae family. Turmeric extract contains major amounts of mineral dyes, curcumin, curcuminoids, phenolic compounds and volatile oils including turmerone, atlantone and zingiberene. Vitale *et al.* (2010) showed that topical application of different levels of hydroethanolic extract of turmeric rhizome remarkably accelerated wound healing activity by increasing in the rate of wound contraction and re-epithelialization, tensile strength value and collagen deposition in rat as an in vivo experimental wound models, and suggested to use various types of wounds in animal and human beings. Curcumin is a phenolic compound which is isolated from *Curcuma longa* Linn and used for its various biological and therapeutic properties. It also has antioxidant, anti-inflammatory, antimutagenic, anticarcinogenic, anti-infective and anticoagulant effects. Curcumin can improve wound healing by its anti-inflammatory, anti-oxidant and anti-infectious properties and also because of the prevention of STAT, TNF- α , cyclin D1, COX-2, NF- κ B, IL (1 β , 6, 8) expressions, and down-regulation of MMP- 8 expression. Curcumin also increases collagen deposition, tissue remodeling, fibroblast proliferation, granulation tissue formation and vascular density. It also prevented the growth of dangerous pathogens like methicillin-resistant *Staphylococcus aureus* (MRSA), *P. gingivalis*, *P. intermedia*, *F. nucleatum*, and *T. denticola*.

Mayana (*Plectranthus scutellariodes* L.) is a common medicinal plants and also used as ornamental plants. The plants contain several phytochemicals such as terpenoids, phenolic compounds, saponins, and anthraquinones, alkaloids, saponin, flavonoids, tannin, volatile oils, and quercetin. Several studies suggest that this plants has antimicrobial, antioxidant, anthelmintic, antimalarial, analgesic, anti-inflammatory, immunostimulant, wound healing, larvicidal, phytoremediative, COX inhibitory, anticancer, antidiabetic, butyrylcholinesterase inhibitory, fungistatic properties. A study on the wound healing activity of ointment formulation of this Mayana plants on inflicted wounds to albino rats showed showed significant wound healing activity with a significant decrease in wound length and size (Stuart, 2021).

On the other hand, *Hagonoy* (*Chromolaena odorata* L.) is also known as the Devil Weed. Hagonoy is a bushy herb or subshrub with long rambling branches, spreading into tangled dense thickets up to 2 meters high. It has become one of the most invasive weeds of Asia and Africa, with consequent economic and ecological burden. This plant is also been reported to be toxic to cattle and can cause allergic reactions. *Hagonoy* has active phytochemical substances which are flavonoid aglycones (flavanones, flavonols, flavones) including acacetin, chalcones, eupatilin, luteolin, naringenin, kaempferol, quercetin, quercetagenin, and sinensetin; terpenes and terpenoids; essential oils; eupolin; alkaloids including pyrrolizidine; saponins and tannins; phenolic acids including ferulic acid and protocatechuic acid; and phytoprostane compound including chromomoric acid. With these active components, several studies proved that *Hagonoy* has antimicrobial, wound healing, hemostatic, antioxidant, anti-inflammatory, wound healing, platelet protective, anticancer, hypoglycemic, hypolipidemic, insecticidal, anti-anemic properties. Some studies on *Hagonoy* extract showed that the molecular mechanisms for its wound healing and hemostatic activities is by promoting the Balb/c 3T3 fibroblast migration and proliferation. Results showed accelerated hemostatic and wound healing activities by altering the expression of genes, including HO-1, TXA, and MMP-9. Another study evaluated the effect of Eupolin extract on hydrated lattice contraction by human dermal fibroblasts, an in vitro model of wound contraction. It was postulated that synergistic properties of components of the natural

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extract contributed to the positive effects on various wound healing mechanisms (Stuart, 2021).

From this information on the effect of Turmeric, *Mayana* and *Hagonoy* on wound healing, the plants can be used as an alternative treatment on animal wounds particularly on Darag native chicken, hence this study was conducted.

Objectives of the Study:

Generally, the study was conducted to evaluate the healing effect of different ethnobotanicals in the wound of capon Darag native chicken cockerels.

Specific Objectives:

1. Evaluate the healing effect of different ethnobotanicals in terms of the decrease in the size of wounds of capon Darag native chicken cockerels.
2. Determine which plant extracts has the least healing period.
3. Compare the healing effect of different plants from the commercial wound solution.

(11) Materials and Methods

Materials

The study utilized a total of forty five heads of Darag native chicken cockerels ages ranging from 2 – 3 months old; rearing pens; commercial feeds; surgical instruments for capon such as surgical blade with holder, forceps, spreader, testes scooper, alcohol, cotton, gauze pad and scissors; fresh plant extracts (turmeric, *mayana*, *hagonoy*), betadine, measuring tape, labeling materials, record book and camera for documentation.

Methods

Research Design and Treatments

The study was arranged in Completely Randomized Design and treatments includes: T1 – Turmeric (*Curcuma longa*) plant extracts, T2 – Mayana (*Plectranthus scutellarioides* L.) plant extracts, T3 – *Hagonoy* (*Chromolaena odorata* L.) plants extracts, T4 – Combined Turmeric, *Mayana* and *Hagonoy* plant extracts and T5 – Control (Betadine).

Research Procedure

Preparation of Experimental Animals. The study was conducted at Animal Science Research Area of CapSU Burias and CAF Laboratory. Experimental animals were raised at the research area for one month before they are subject for surgical procedures. Draw lots scheme were employed in assigning the birds for their treatments. Proper animal health procedures were practiced in the birds such as deworming and vaccination. Commercial feeds were used as feeds of the chicken for the entire duration of the study.

Preparation of Plant Extracts. All plants were gathered from the nearby areas in the study sites. Fresh leaves of *Mayana* and *Hagonoy* were cut and separated from its stem, washed thoroughly, drained to remove some water and pounded to get the liquid extracts. While

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Turmeric were also washed, pounded and squeezed to get its extract. All plant extracts were stored in a clean spray bottle and labeled.

Caponization of Experimental Birds and Application of the Plant Extract. Prior to the start of the study period, legal papers on the use of experimental animals were secured and with the assistance of veterinarians. The birds were fasted for 16 hours before the surgical procedures. This is to avoid excessive bleeding and to increase the prominence of the testes during surgery. Proper use of the PPE, disinfection of the materials needed, tables, as well as disinfection of the incision site was done. Each cockerel was incised of 2 cm both in their right and left upper abdomen (position where the testes is situated). Using the spreader, forceps and scooper, the testes were exposed and were cut immediately. After testes removal, each incision wounds of the chickens were sprayed by the different plant extracts and commercial wound spray based on their assigned treatments.

Data Gathered. Birds were put in their assigned cages and were observed for the changes in the healing effect of the plants and commercial wound solution for 15 days. Wound measurement were done on the 3rd, 6th, 9th, 12th and 15th day post surgical operation. Administration of the plant extracts and commercial wound solution was done everyday for 7 days.

Data Analysis

Results of the study was analyzed using Analysis of Variance (ANOVA). Differences among treatment means were compared using the Least Significant Difference (LSD) and results interpreted at five percent level of significance..

(12) Results and Discussions

Healing Effect of Ethnobotanicals Extracts on Wounds of Capon Darag Native Chicken Cockerel

The effect of Turmeric, *Mayana* and *Hagonoy* extracts in the decreased in the size of wound diameter incision is shown in Table 1.0.

Data revealed that after 3 days post surgical operation, the incision diameter was decreased from 1.38 cm to 1.61 cm by which the cockerels that were applied with commercial wound solution (T5) obtained the lowest diameter decreased with a mean of 1.38 cm. This was followed by the cockerels applied with pure *Hagonoy* extracts (T3) and combined plants extracts (T4) with a mean of 1.45 cm and the birds applied with Turmeric (T1) with a mean of 1.61.

After 6 days post surgical operation, the study revealed that cockerels applied with commercial wound solution (T5) obtained the smallest wound incision size of 0.73cm, followed by the *Hagonoy* extract (T3) and mix plants extracts (T4) with wound size diameter of 0.88 cm, turmeric extract (T1) with a mean of 0.95 cm, and *Mayana* extracts (T2) with a mean of 1.02 cm.

On the 9th day of post surgical operations, cockerel applied with *Hagonoy* extracts (T3) obtained the smallest wound size diameter (0.09 cm), followed with the cockerels given with Turmeric (T1-0.13 cm), *Mayana* (T2-0.13 cm), and commercial wound solution (T5-0.21 cm). Birds applied with mix plants (T4) obtained the biggest wound incision size of 0.46 cm.

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After 12th days of post surgical operation, capon cockerels applied with *Mayana* (T2) and *Hagonoy* (T3) plant extracts were totally healed of their wounds. This data is lower than the average number of days of wound healing which is generally 28-42 days (4 to 6 weeks) according to Wallace *et al.*, (2023). Moreover, birds given with Turmeric (T1), mix plants extracts (T4) and commercial wound solution (T5) still had wounds with a size diameter of 0.03 cm, 0.07 cm and 0.21 cm, respectively. This numeric difference in number of days of wound healing, however, was not statistically significant.

On 15th day of post capon operation, both Turmeric (T1), *Mayana* (T2) and *Hagonoy* (T3) plants extracts completely healed the cockerels' wounds. On the other hand, cockerels applied with commercial wound solution (T5) and mix plants extracts (T4) still has a very minimal wound size of 0.03 cm and 0.06 cm, respectively.

Analysis of variance (ANOVA) showed that Turmeric, *Mayana*, and *Hagonoy* extracts has a comparable effect with the commercial wound solution on healing of wounds of capon Darag native chicken cockerel at different time frames. This means that all plants extracts can be an alternative solution to the commercially available brand.

The wound healing effect of the above-mentioned plants was because of their active phytochemical components such as curcumin for Turmeric which can improve wound healing by its anti-inflammatory, anti-oxidant and anti-infectious properties and also because of the prevention of STAT, TNF- α , cyclin D1, COX-2, NF- κ B, IL (1 β , 6, 8) expressions, and down-regulation of MMP- 8 expression. Curcumin also increases collagen deposition, tissue remodeling, fibroblast proliferation, granulation tissue formation and vascular density (Vitale *et al.* 2022). The Eupolin extract from *Hagonoy* which affects the hydrated lattice contraction by human dermal fibroblasts, an in vitro model of wound contraction. It was postulated that synergistic properties of components of the natural extract contributed to the positive effects on various wound healing mechanisms (Stuart, 2021). Flavonoids, and terpenoids from *mayana* are also reported to promote wound healing primarily due to their astringent and antimicrobial property, which could be responsible for wound contraction and improved rate of epithelialization (Asumang *et al.*, 2021).

Table 1.0 Healing effect of Ethnobotanical plant extracts on the wound of capon Darag native chicken cockerel.

TREATMENTS	POST SURGICAL (CAPON) DAYS				
	3 rd <i>ns</i>	6 th <i>ns</i>	9 th <i>ns</i>	12 th <i>ns</i>	15 th <i>ns</i>
T ₁ - Turmeric Plant Extract	1.61	0.95	0.13	0.03	0.00
T ₂ - Mayana Plant Extract	1.55	1.02	0.13	0.00	0.00
T ₃ - Hagonoy Plant Extract	1.45	0.88	0.09	0.00	0.00
T ₄ - Combined Plant Extract	1.45	0.88	0.46	0.09	0.06
T ₅ - Commercial Wound Solution	1.38	0.73	0.21	0.07	0.03

ns – not significant 0.00 – wounds are totally healed cv – 19.82

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(13) Conclusions and Recommendations

Based on the findings of the study, the researchers drawn the following conclusions:
The use of Turmeric, *Mayana* and *Hagonoy* extracts can facilitate the healing of the wounds of capon Darag native chicken cockerel faster by decreasing the wound size and diameter. *Mayana* and *Hagonoy* extracts completed the healing of wounds at 12 days of post capon operation. Turmeric, *Hagonoy* and *Mayana* extracts has a comparable healing effect with the commercial wound solution.

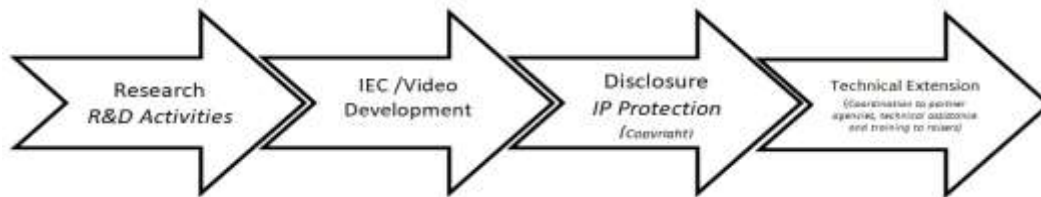
The researchers recommend the use of *Hagonoy*, *Mayana* and Turmeric as alternative wound treatment for chickens. Conduct further studies and development of these plants such as oil and ointment. Shelf life and laboratory analysis on its chemical components should also be conducted.

(14) Output

EXPECTED OUTPUTS	DETAILS
People Services	Train farmers on the proper use of the plant extracts as alternative treatment of their animals wounds/ incision.
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials for the use of Turmeric, <i>Mayana</i> and <i>Hagonoy</i> plants as alternative source for wound treatments.
Patents/IP	File copyrights for the developed IEC materials. File patent/UM for the product to be developed.
Places and Partnerships	Forge linkages and partnerships to LGU's, DA RFO 6, Capiz PVO, and other organizations.
Products	Turmeric, <i>Mayana</i> and <i>hagonoy</i> wound solution
Impact to Community	Increase awareness of the community particularly on the farmers for the uses of Turmeric, <i>Mayana</i> and <i>Hagonoy</i> as wound solution. Lessen production cost of raisers for they will no longer used the commercial wound solution.

(15) Technology Transfer Activity Conducted

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(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments

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